DATA ANALYTICS TRACK / PHASE 2

PulseIQ Phase 2: People and Teams

Aditya Mergurwar | Data Analytics Track | Phase 2

Repository: https://github.com/adityamergurwar/pulseiq/tree/phase-2
Git tag: phase-2
Live dashboard: https://pulseiq-workforce.streamlit.app

Build summary

Five connected dashboard modules extend the existing workforce analytics foundation: attendance and leave, attrition, team delivery, payroll cost, and budget/KPI monitoring. Governed DuckDB views supply consistent definitions to the Streamlit pages. The build adds calendar-aware attendance context, a clearly separated grouped risk signal, sprint-level delivery measures, privacy-safe payroll aggregates, and signed budget variance with specific follow-up actions.

Five task outcomes and evidence

Task	Status	Evidence
6 - Attendance and leave	**COMPLETE**	Attendance rate, holidays, leave type and unusual-month findings Validation detail: Phase 2 checks
7 - Attrition	**COMPLETE**	Observed exits, grouped risk, tenure and reasons Validation detail: Phase 2 checks
8 - Team productivity	**COMPLETE**	Team/sprint metrics, rules, health comparison and manual checks Validation detail: Phase 2 checks
9 - Payroll cost	**COMPLETE**	Cost trend, privacy-safe groups, exclusions and reconciliation Validation detail: Phase 2 checks
10 - Budget and KPI	**COMPLETE**	Signed variance, traffic lights, targets and recommendations Validation detail: Phase 2 checks

All five Phase 2 tasks have working local pages and validation evidence. The validation note contains the hand-check values, reference reconciliation and test summary.

VERIFICATION

Evidence and outcomes

18	18 / 18	28 / 28	0
automated tests passed	payroll reference months match	metric views populated	declared warehouse orphans

The latest acceptance run covers 29 source tables with no unexplained row-count differences; all 28 governed metric views return results. Independent checks confirm the scheduled-day attendance denominator, two team/sprint calculations, three budget variances and the complete PQ-09 reference comparison.

Task-level proof

Area	Verified result
Attendance	Lowest company rates: May 2026 92.89% (1 holiday, 545 approved leave days); January 2026 93.29% (3 holidays, 573 leave days); April 2026 93.35% (2 holidays, 405 leave days). The page marks holiday context without claiming it caused the rate change.
Attrition	September 2026: 3 observed exits (2 voluntary, 1 other); 0.70% monthly attrition. The baseline risk view is separate and grouped; small groups are suppressed.
Teams	SP-01 manual checks match the metric views. T01: 162 points, 4.64-day cycle, 96.00% on-time, 90.91% complete, 2 blocked, 4 reopened. T02: 210 points, 4.79-day cycle, 93.65% on-time, 92.65% complete, 2 blocked, 4 reopened.
Payroll	PQ-09: 18 of 18 reference months match; the largest clean payroll-to-ledger difference is INR 0.00000004 against an INR 0.01 tolerance. Four excluded rows are two duplicate run keys and two unmatched employee references; the app lists row and reason without personal identifiers or pay amounts.
Budget and KPI	September 2026 independent variances: D01 +INR 214,404.24 (+1.03%, amber), D03 +INR 30,889.62 (+0.48%, amber), D02 -INR 42,581.21. D01 and D03 are the two largest overspends and each has a review-open-roles/overtime action. KPI status follows each target's direction.

Learning notes

- Calendar context should accompany attendance rates before a month is interpreted.
- Observed exits and predicted risk answer different questions and should remain separate.
- Velocity needs cycle time, on-time delivery, blocked work and reopen counts for a fair team view.
- Reconciliation proves cost totals; a written variance sign makes the action clear.

Issues, fixes and next phase

Four invalid payroll rows were traced to duplicate run keys or unmatched employee references; clean reporting excludes them with row-level reasons while preserving the raw source. Attendance dips now include leave and holiday context, and the team page balances velocity with delivery-quality measures. Next, extend the governed model to revenue, cash and AI signals, and connect refreshes to new source data.

DASHBOARD EVIDENCE

Attendance and leave

Live page evidence showing monthly attendance, public-holiday markers, approved leave, late arrivals and overtime. Task-specific code and verification links are available in the Phase 2 table on page 1.



Dashboard: https://pulseiq-workforce.streamlit.app | Repository and Phase 2 release: https://github.com/adityamergurwar/pulseiq/tree/phase-2